

8086 CPU Architecture & Internal Registers

****Q.** Explain the internal architecture of the 8086 microprocessor. Describe the functions of the Bus Interface Unit (BIU) and the Execution Unit (EU).

> ****Definition to Remember****

1. Bus Interface Unit (BIU)

The BIU handles all data and address transfers on the system buses.

2. Execution Unit (EU)

The EU is responsible for decoding and executing the fetched instructions.

3. Parallel Operation

Because the BIU and EU operate independently, the BIU continuously fills the 6-byte queue from memory while the EU executes the previously fetched instructions. This prevents the EU from sitting idle waiting for memory fetches, drastically reducing execution time.

> ****Must-Write Points (for 10 marks)****

> ****Quick Recall****